Homework

DM9 - Forming a committee - 45 min, after S35.

A team consists of five teachers and seven students, all distinct. four people.

1. How many different committees can be formed?

2. How many committees contain exactly two pupils?

3. How many committees contain at least one teacher and at least one pupil?

4. From the members of a committee in question 2, we appoint a teacher as chair and a student secretary. Count the possible choices of committee together with these two roles.

5. By counting the committees according to their number of teachers, prove that

$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4}.$$